

PLS-SEM Reporting Checklist

Every required output for a defensible PLS-SEM manuscript, mapped to Hair et al. thresholds.

STEP 1 REFLECTIVE MEASUREMENT MODEL

METRIC	THRESHOLD	IF IT FAILS
☐ Indicator loadings	≥ 0.708	Consider dropping items with loadings < 0.40
☐ Cronbach's alpha	$0.70 - 0.95$	Below 0.60 = inadequate; above 0.95 = redundancy
☐ Composite reliability ρ_C	$0.70 - 0.95$	Report ρ_A as well (Dijkstra-Henseler)
☐ AVE (convergent validity)	≥ 0.50	Drop weakest loader; check theoretical fit
☐ HTMT (discriminant validity)	< 0.85 strict / 0.90 lib.	Report bootstrap CI; must exclude 1.0
☐ Fornell-Larcker criterion	$\sqrt{\text{AVE}} > \text{inter-construct } r$	HTMT is preferred; report both if space
☐ Cross-loadings	Within-construct $>$ cross-load	Locate the offending indicator pair

STEP 2 FORMATIVE MEASUREMENT MODEL (if applicable)

METRIC	THRESHOLD	IF IT FAILS
☐ Convergent validity (redundancy)	Path ≥ 0.70	Requires a global single-item proxy
☐ Indicator collinearity (VIF)	< 3 (tolerant ≤ 5)	Merge or drop redundant indicators
☐ Indicator weights (bootstrapped)	Significant at $p < 0.05$	Retain non-sig. weights if loading ≥ 0.50
☐ Indicator loadings (retention)	≥ 0.50 to keep	Below 0.50 with non-sig weight $\rightarrow$ remove

Skip this section if all constructs are reflective.

STEP 3 STRUCTURAL MODEL

METRIC	THRESHOLD	IF IT FAILS
☐ Path coefficient collinearity (VIF)	< 3 (tolerant ≤ 5)	Merge overlapping predictors
☐ Path coefficients + significance	p-values via 5,000 bootstraps	Report BCa 95% CI, not just p
☐ Coefficient of determination R^2	$0.75 / 0.50 / 0.25$	Substantial / moderate / weak
☐ Effect size f^2	$0.02 / 0.15 / 0.35$	Small / medium / large
☐ Predictive relevance Q^2	> 0 via blindfolding	Or use $PLSpredict$ (Shmueli et al.)
☐ Model fit (SRMR)	< 0.08 (< 0.10 acceptable)	Report d_{ULS} , d_G bootstrap CI

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STEP 4 REPORTING ESSENTIALS (methods section must-haves)

METRIC	THRESHOLD	IF IT FAILS
☐ Software + version	AnalyVa 4.2.2 / SmartPLS 4	<i>Cite it (see analyva.com/cite)</i>
☐ Bootstrap settings	5,000 subsamples · BCa	<i>State one- vs two-tailed a priori</i>
☐ Random seed	Report for reproducibility	<i>AnalyVa exports the seed automatically</i>
☐ Sample size + justification	Inverse sqrt / G*Power	<i>Power for smallest expected effect</i>
☐ Missing data treatment	Mean / EM / listwise	<i>Report % missing per indicator</i>
☐ Common method bias	Harman's / marker variable	<i>Add procedural remedies to design</i>
☐ Model specification	Reflective vs formative	<i>Justify each choice theoretically</i>

STEP 5 ADVANCED ANALYSES (report if performed)

METRIC	THRESHOLD	IF IT FAILS
☐ Mediation (indirect effects)	VAF / Zhao et al.	<i>Bootstrap indirect effect CI</i>
☐ Moderation (interaction)	Two-stage or orthog.	<i>Report simple slopes plot</i>
☐ MGA (multi-group analysis)	MICOM invariance first	<i>Then permutation / Henseler MGA</i>
☐ IPMA (importance-performance)	Rescale on 0–100 scale	<i>Importance vs performance chart</i>
☐ FIMIX / PLS-POS segmentation	AIC3 / EN / BIC for k	<i>Report profile of each segment</i>
☐ Endogeneity (Gaussian copula)	Ramsey RESET / Park-Gupta	<i>Correct via Gaussian copula</i>

HOW TO USE THIS CHECKLIST

Print it. Keep it beside your keyboard while you write. Tick each box as you write the corresponding sentence in the methods or results section. If a box stays empty, you owe your reviewer an explanation for why — or you owe your model an extra run.